INCH-POUND
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

TRUCK, LIFT, FORK, DIESEL-ENGINE-DRIVEN, PNEUMATIC-TIRED, 30,000-POUND CAPACITY AT 24-INCH LOAD CENTER

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This commercial item description (CID) covers the general requirements for a diesel engine driven, front-wheel-drive, rear-wheel-steer, pneumatic-tired, 30,000-pound at 24 inch load-center capacity, forklift truck. The forklift is intended for handling cargo in and around warehouses, loading platforms, docks, and on paved or other hard surfaces.

2. SALIENT CHARACTERISTICS.

2.1 Safety. The operator driven forklift shall comply with ANSI/ITSDF B56.1, Underwriters Laboratories (UL) 558 Type DS, and Occupational Safety and Health Administration (OSHA) standards in effect at the time of manufacture. An operator's seatbelt conforming to SAE J386, and restraint devices designed to ensure the operator's upper body remains entirely within the protection of the overhead guard in the event of tip-over shall be provided.

2.2 Environmental requirements. The engine shall start without preheating within 5 minutes and the truck shall operate as specified herein within 15 minutes after engine start, in any ambient temperature from 0 to 120 °F. The truck, with type A winterization system, shall be capable of operating in temperatures ranging from -65 to 125 °F. The truck, with type C winterization system, shall be capable of operating in temperatures ranging from -25 to 125 °F.

2.3 Engine. The forklift shall be powered by a standard, commercial, turbocharged diesel engine. The engine shall operate on diesel fuel in accordance with ASTM D975 without detrimental effects on the engine or its performance.

2.4 Fuel system. The forklift shall have all necessary filters, water separators, and components required for diesel and biodiesel fuel operation. The forklift shall have a fuel tank of sufficient capacity to allow at least eight hours of continuous operation.

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AMSC N/A FSC 3930

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2.5 Starting system. The engine starter switch shall not activate the engine starter while the engine is running, or while the transmission is in forward or reverse gear. The starter switch shall only operate when the transmission is in the “neutral” or “park” positions.

2.6 Transmission. A torque converter and a power shift type of transmission shall be furnished with at least two forward speeds and two reverse speeds. The transmission shall provide for positive inching control or declutch control that permits lifting of rated load at maximum lifting speed while the transmission control is in either the forward or reverse direction without vehicle movement. This control shall also apply to the service brakes.

2.6.1 Transmission oil cooler. When specified (see 6.3), a transmission oil cooler shall be furnished.

2.7 Hydraulic system. A pressure-relief protection device is required along with pump(s), cylinders, control valves, control lever, filter(s), reservoir, hoses, and all other components necessary to make a complete hydraulic system.

2.8 Dimensions. The overall truck height, with mast collapsed, shall not exceed 143 inches. The overall width of the truck, at the widest point, shall not exceed 102 inches. The maximum length, including the forks, shall not exceed 268 inches.

2.9 Wheelbase. The wheelbase of the truck shall not exceed 130 inches.

2.10 Uprights and carriage. With no load on the forks, the seated operator, with seatbelt engaged, shall be able to see at least one fork tip at all lift heights and fork positions. The carriage shall have a minimum width of 99 inches.

2.11 Forks and carriage. The forks shall be mounted so that when fully lowered and forklift operating in reverse, there is no damage or degradation to the fork tines, their attachment, or any other component of the carriage, including fork positioner and side shift.

2.11.1 Standard forks. The fork tine dimensions shall be:

a. Length: 96 ±1 inches
b. Width: 8 inches
c. Thickness: 3.5 inches

2.11.2 Optional forks. When specified (see 6.3), the fork tine dimensions shall be:

a. Length: 114 ±1 inches
b. Width: 8 inches
c. Thickness: 3.5 inches

When these forks are specified, the forklift shall be rated to handle a load of 21,000 lb. at a load center of 57 inches.

2.12 Fork positioner. A hydraulically-powered fork positioner shall be furnished and controlled by the operator while seated. The fork positioner shall allow tine spacing of 3 inches (or less) to 95 inches (or more), measured between centerlines of the forks.

2.13 Side shift. Hydraulically-powered side shift capability shall be furnished. It shall be controlled by the seated operator, with at least 6 inches of side shift capability each side of center (12 inches minimum travel) with rated load on the forks.

2.14 Load backrest. The load backrest, in conjunction with the forks and carriage, shall provide a vertical, rear-load guard of at least 48 inches high, measured from the load carrying surfaces of the forks.

2.15 Steering. Power steering, with emergency steering in the event of power failure, shall be furnished.

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2.16 Service brakes. Service brakes conforming to ANSI/ITSDF B56.1 shall be furnished, except maximum brake-pedal force during stopping distance and drawbar drag tests shall not exceed 80 lbs.

2.17 Parking brake. Parking brake shall conform to ANSI/ITSDF B56.1.

2.18 Electrical system. A 12-volt, direct-current electrical system shall be furnished. The electrical system shall consist of all electrical components necessary for operation of the forklift. The forklift shall have sufficient electrical grounding to prevent static discharge.

2.19 Instruments and controls. In addition to the instruments supplied on the standard commercial forklift, an hour meter and tachometer shall be installed in the instrument panel. Except for the hour meter, all instruments shall illuminate. All load-motion controls shall be right-hand operation controls and of the self-centering type, i.e., controls shall return to the neutral position when released.

2.20 Lighting. A minimum of four sealed beam floodlights shall be furnished; two in front facing forward and two mounted in the rear facing rearward. Tail light(s) and brake stop light(s) shall be installed. All lights shall be protected by location placement or guards. Individual operator controlled switches shall be provided for front floodlights, rear floodlights, and adjustable lights.

2.21 Horn. A horn shall be installed, with the horn actuator located in the center of the steering wheel.

2.22 Back-up alarm. An audible back-up alarm shall be installed on the forklift. The back-up alarm shall automatically activate when the transmission selector is placed in reverse. The back-up alarm must conform to SAE J994.

2.23 Operator's overhead guard. The overhead guard shall be a falling-object protective structure (FOPS) conforming to ANSI/ITSDF B56.1. Overhead guard height from the ground to the top of the overhead guard shall not be more than 120 inches.

2.24 Cab. The forklift shall be equipped with an enclosed cab that shall incorporate the FOPS requirements of ANSI/ITSDF B56.1. The cab shall have a door on each side of the vehicle with at least one window per door, which can be opened for ventilation. The windows shall be capable of being locked in either the open or closed positions. In addition to the door windows, the cab shall be furnished with a front windshield, rear window, and roof window. The windshield and all windows shall be safety glass. The cab interior shall be furnished with water-resistant, non- absorbent thermal insulation. The interior shall come equipped with a floor mat.

2.24.1 Extinguisher. A commercial fire extinguisher shall be mounted on the truck in an easily accessible location. The fire extinguisher shall be a minimum 2-1/2-lb capacity, ABC type, or equivalent. The fire extinguisher shall be UL listed.

2.24.2 Heater, defroster, and air conditioner. The manufacturer's heaviest duty heater and defroster shall be furnished. The cab shall come equipped with an air conditioner.

2.24.3 Seat. The cab shall have a vinyl, mechanical-suspension seat with an operator presence system.

2.24.4 Handrail. The cab shall be installed with handrails to assist with operator entry and exit.

2.25 Windshield wipers. There shall be at least three wipers furnished; one on the front windshield, one on the rear window, and one on the roof window.

2.26 Mirror. The truck shall be equipped with an adjustable rearview mirror mounted within the operator’s field of view as described in SAE J985. The exact mounting location shall be determined by the manufacturer and shall increase the driver’s field of vision. The truck shall come equipped with adjustable side-view mirrors.

2.27 License plate. The truck shall come equipped with a rear, license-plate mounting bracket with plate-attaching hardware.

2.28 Performance requirements. The truck shall be capable of meeting the following performance

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requirements:

a. Lifting speed. Lifting speed shall be at least 81 feet per minute without load and 71 feet per minute with rated load.

b. Lowering speed. Lowering speed shall be at least 98 feet per minute with rated load on the forks and 94 feet per minute without load.

c. Turning radius. The maximum turning radius of the truck, measured to the outer tire, shall not exceed 180 inches.

d. Travel speed. Travel speed shall be 16 miles per hour (mph) maximum, in both forward and reverse, while carrying rated load on the forks. The vehicle speed shall be governed so that the maximum speed is not exceeded for safe operations.

e. Slope ascension. The truck shall be able to accelerate up a 34 percent slope at 1 mph with rated load on the forks.

f. Upright tilt. The mast shall have at least 15 degrees of forward tilt and at least 12 degrees of rear tilt without a load on the forks.

g. Upright height (collapsed mast height). The dimension from ground to top surface of the uprights with uprights vertical, shall not exceed 215 inches with no load on the forks.

h. Maximum forklift height. The maximum forklift “lift height” shall be at least 148 inches with rated load on the forks and the forks horizontal when measured from ground to top surfaces of the forks.

i. Ground clearance. With rated load on the forks and mast vertical, ground clearance beneath the mast shall be at least 7 inches. Ground clearance at the center of the wheel base with rated load shall be a minimum of 14 inches.

j. Load drift. With hydraulic fluid at normal operating temperature, the lift assembly shall hold rated load at maximum lift height for at least 10 minutes with not more than 1.75 inches of vertical drift, and not more than 1 degree of rotational drift.

k. Stability. The forklift shall meet the ANSI/ITSDF B56.1 "forward stacking," "forward travel," "lateral stacking," and "lateral travel" stability requirements.

l. Noise limits. Maximum allowable noise level shall not exceed 84 decibel A (dB(A)) at the operator's ear. If the noise level at the operator's ear is between 85 to 92 dB(A), a permanent warning decal or placard shall be installed inside the cab clearly visible to the operator. Noise level shall not exceed 92 dB(A).

2.29 Painting. The forklift shall be primed and painted using standard commercial practices. The final paint shall be a standard, commercially-available polyurethane. The paint finish shall show no evidence of uneven application, curtains, runs, snags, orange peel, lack of adhesion, or other defects. The color shall be the manufacturer’s standard color. When specified (see 6.3), the manufacturer shall make color 24052 of SAE-AMS-STD-595 available.

2.30 Identification plate. A corrosion-resistant identification plate, hydraulic schematic wiring schematic, and all warning and instruction plates shall be provided and mounted on the vehicle. The identification plate shall contain the following information:

a. Nomenclature.
b. Manufactures Name
c. Manufacturer’s Address
d. Manufacturer’s Phone Number.
e. Manufacturer’s Model Number.
f. Manufacturer’s Serial Number.

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g. Registration Number.
h. National Stock Number.
i. Vehicle Curb Weight.
j. Payload, Maximum (lbs).
k. Gross Weight, Maximum (lbs).
l. Date of Delivery.
m. Warranty (months/miles).
n. Contract Number.
o. Ship to_____ Mark for _____.
p. US Property.

2.31 Lubrication data plate. A permanently-marked, corrosion-resistant plate marked with all lubrication fittings and components and identifying the type and grade of lubricant required for all operational temperatures shall be provided and mounted on the vehicle.

2.32 Transportation data plate. A permanently-marked, corrosion-resistant plate marked with transportation data shall be provided and mounted on the vehicle. The transportation plate shall include the following information:

a. Side and rear silhouette views of the forklift
b Horizontal and vertical location of the center of gravity of the forklift on the silhouette view
c. Shipping weight
d. Loading cubage
e. Overall height, width, and length
f. Lift and tie-down information

2.33 Manuals. The truck shall come equipped with an operator's, parts, and service manual. The parts manual shall list the part numbers for contractor-manufactured parts, plus the part numbers for purchased sub-level components and whole components. Manuals shall be provided in accordance with DFARS 252.227-7015, *Technical data-Commercial Items*.

2.34 Walkway coating. Floor plates and step surfaces shall be coated with a nonslip coating compound, or be furnished with a nonslip metal or tread-plate surface.

2.35 Lifting and tie-down provisions. There shall be permanently-installed lifting devices that enable the forklift to be lifted in its normal travel configuration, and tie-down devices that enable the forklift to be tied down to the floor of the transportation media. All devices shall have an inside diameter of 3 inches and a minimum structural safety factor of 2:1, based on static load. The notation "LIFT HERE" and "TIEDOWN" shall be stenciled, in black, near each lifting and tie-down device.

2.36 Electromagnetic interference (EMI). The forklift shall comply with EMI requirements of SAE J551-1 and SAE J551-2.

2.37 Highway transportability. The truck, when loaded on a semitrailer, shall be within the highwaypermit limits for all states.

2.38 Exhaust system. The truck shall contain all of the standard commercial components composing an exhaust system. When specified (see 6.3), the exhaust system shall include a commercially available spark-arrestor muffler. The spark-arrestor muffler shall be sized to meet requirements for engine exhaust flow at rated engine speed.

2.39 Winterization.

2.39.1 Type A winterization. When specified (see 6.3), the truck shall be furnished with type A winterization that protects to -65 °F and shall consist of the following:

a. Power-plant heaters. Engine coolant, engine oil, and battery heaters shall be provided. All heaters shall operate on either 110 volts @ 60 Hz or 220 volts @ 50 Hz alternating current (VAC). A three-wire, 25-feet long, weatherproof cable of adequate capacity for all heaters being used

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simultaneously shall be provided. The cable shall be able to connect to NATO and American connectors. A stowage place shall be provided on the truck to keep the cable in when not in use. The heaters shall be as follows:

(1) The coolant heater shall be installed in the engine block or lower coolant inlet hose. A coolant-circulating pump, driven by a 110 or 220 VAC motor shall be provided when a coolant inlet hose heater is furnished. The heater shall have adequate capacity to maintain engine coolant at a temperature of 10 °F in an ambient temperature of -65 °F. It shall be controlled to limit engine coolant to not more than 150 °F.

(2) An engine oil heater with adequate capacity to maintain engine oil at a temperature of atleast 10 °F in an ambient temperature of -65 °F shall be furnished. It shall be controlled to limit engine oil temperature to not more than 150 °F.

(3) A battery heater shall be provided. It shall have adequate capacity to maintain batteryelectrolyte at a temperature of at least 10 °F in an ambient temperature of -65 °F, and shall have a thermostat to limit temperature of the electrolyte to not more than 80 °F.

b. Cab. The cab shall meet the requirements of 2.24.2 and shall also contain thermal insulation, if necessary, to meet the heating requirements of 2.39.1.

c. Cab heater and defroster. The heater shall be of sufficient capacity to maintain a temperature of 40 °F at cab floor level in an ambient temperature of -40 °F.

d. Starting aid. Glow plug(s) or a measured-shot, ether injection system shall be provided to assist in engine cold-temperature starting.

e. Antifreeze. Engine coolant shall be protected to -65 °F with antifreeze.

2.39.2 Type C winterization. When specified (see 6.3), the truck shall be furnished with type C winterization that protects to -25 °F and shall consist of the following:

a. Starting aid. Glow plug(s) or a measure-shot, ether injection system shall be furnished.

b. Cab. The cab shall meet the requirements of 2.24.2 and shall also contain thermal insulation, if necessary, to meet the heating requirements of 2.39.2.

c. Cab heater and defroster. The heater shall have sufficient capacity to maintain a temperature of at least 40 °F at cab floor level in an ambient temperature of -20 °F.

d. Antifreeze. Engine coolant shall be protected to -25 °F with antifreeze.

2.40 Tires. The tires shall be pneumatic, and tire loading shall not exceed those specified in the Tire and Rim Association (TRA) Yearbook.

2.41 Cybersecurity. The forklift shall be designed, produced, and delivered with cybersecurity considerations in accordance with DoDI 8500.01, DoDI 8510.01, and AFI 17-101 and referenced security controls for applicable Information Technology (IT) and/or Operational Technology (OT), commonly referred to as Platform Information Technology (PIT). Commercial policy and standards, such as SAE J3061, SAE J3101, and various ISO/IEC standards may be used in lieu of DoDI/AFI, as applicable.

2.41.1 Secure communication. The forklift shall be designed, produced, and delivered with the bare minimum of any wireless communication (i.e. cellular, Wi-Fi, telematics, bluetooth, etc.) while still meeting operational/performance specifications within this CID. Any wireless communications remaining within the final configuration of the forklift shall be secured to mitigate known vulnerabilities and threats in accordance with applicable policy and standards as referenced in paragraph 2.40 All risks associated with known vulnerabilities and potential threats of any wireless communication capability shall be identified, assessed, and reported.

2.41.2 Hardware supply chain integrity. The forklift shall be designed, produced, and delivered with security controls for the supply chain integrity of electronics hardware (i.e. engine control units (ECUs)) that involve

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memory storage, data processing, and/or interfaces with the communication system or controller area network (CAN bus) within the vehicle. The forklift shall be produced with mitigation to the risk of counterfeit electronic parts being installed in end items or otherwise entering the United States Air Force (USAF)/Department of Defense (DoD) inventory. The contractor shall provide products and components, or sub-components, sourced through original equipment manufacturers or authorized resellers.

2.41.3 Software/firmware supply chain integrity. The forklift shall be designed, produced, and delivered with security controls for the supply chain integrity of all delivered software and firmware. All software and firmware shall function as intended, and shall be free of vulnerabilities (i.e. defects, bugs, malware, etc.), either intentionally or unintentionally inserted as part of the software/firmware throughout the lifecycle.

3. REGULATORY REQUIREMENTS

3.1 Recovered, recycled, or biobased materials. The offeror/contractor is encouraged to use recovered, recycled, bio-based materials to the maximum extent practicable, in accordance with Federal Acquisition Regulation (FAR), part 23.403.

4. PRODUCT CONFORMANCE PROVISIONS

4.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5. PACKAGING

5.1 Packaging. Unless otherwise specified (see 6.3), the preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc. This involves shipment from the supply source to the receiving activity, and reshipment from the receiving activity. The preservation, packaging, and packing shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities, labeling, and marking shall be as specified in the contract or order.

6. NOTES

6.1 Intended use. The forklift truck is intended for use in handling 30,000-lb capacity cargo in and around warehouses, loading platforms, docks, and on paved or other hard surfaces.

6.2 Source of documents.

6.2.1 Air Force Instructions (AFI). Copies of AFIs are available online at https://www.e-publishing.af.mil/.

6.2.2 American National Standards Institute/Industrial Truck Standards Development Foundation (ANSI/ITSDF) standards. Copies of ANSI/ITSDF standards may be obtained from the ITSDF, Suite 460, 1750 K Street NW, Washington, DC 20006, or online at https://www.itsdf.org/.

6.2.3 American Society for Testing and Materials (ASTM) documents. Copies of these documents may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959, or online at https://ww.astm.org.

6.2.4 Department of Defense Instructions (DoDI). Copies of DoDIs are available online at https://www.esd.whs.mil/DD/.

6.2.5 DFAR and FAR. Electronic copies of the DFAR and FAR may be obtained from https://www.acquisition.gov.

6.2.6 Government documents. Copies of specifications, standards, and handbooks are available online at https://assist.dla.mil.

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6.2.7 Occupational Safety and Health Association (OSHA). Copies of OSHA documents may be obtained from Occupational Safety and Health Association, 200 Constitution Ave., NW, Washington, DC 2010, or online at https://www.osha.gov/.

6.2.8 Society of Automotive Engineers (SAE) standards. Copies of SAE standards may be obtained from the Society of AutomotiveEngineers, 400 Commonwealth Drive, Warrendale, PA 15096, or online at https://www.sae.org/.

6.2.9 Tire and Rim Association (TRA) standards. Copies of TRA standards may be obtained from the Tire and Rim Association, Inc., 175 Monroe West Avenue, Suite 150, Copley, OH 44321, or online at https://www.us-tra.org/publications.html.

6.2.10 Underwriters Laboratories (UL) standards. Copies of UL standards may be obtained from the Underwriters Laboratories,Inc., 333 Pfingsten Road, Northbrook, IL 60062, or online at https://www.ul.com/.

6.3 Ordering data. Acquisition documents shall specify the following information:

a. Title, number, and date of this CID.
b. Part identification number (PIN).
c. Transmission oil cooler, when specified (see 2.6.1).
d. Optional forks, when specified (see 2.11.2).
e. When paint is other than manufacturer's standard color (see 2.29).
f. Spark-arrestor muffler, if required (see 2.38).
g. Type of winterization, if any (see 2.39).
h. Packaging requirements, if different (see 5.1).

6.4 Key words.

Front-wheel drive
Rear-wheel steer
Handling cargo
Warehouses
Fork positioner
Load backrest
Cab

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

MILITARY INTERESTS

Custodian:
Air Force - 184

CIVIL AGENCY COORDINATING ACTIVITY

Preparing Activity:
Air Force - 184

Agent:
Air Force - 110

(Project: 3930-2022-001)

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